Feature Report: Time Features

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Basic Information

- **Total Customers**: 61,410
- **Total Features**: 25
- **Feature Columns**: txn_hour_mode, after_hours_txn_pct, after_hours_flag, weekend_txn_count, weekend_txn_pct, weekend_heavy_flag, midnight_txn_count, early_morning_txn_count, late_night_txn_count, midnight_txn_pct, unusual_hours_flag, time_between_txn_mean, time_between_txn_median, time_between_txn_min, time_between_txn_max, time_between_txn_std, very_short_time_between_txn, rapid_fire_txn_count, rapid_fire_flag, txn_hour_std, consistent_timing_flag, business_hours_txn_pct, non_business_hours_heavy, months_active, txn_per_month

Data Quality Checks

☑ No Null Values

All features have complete data.

☑ No Duplicate Customer IDs

Feature Statistics

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero %
`txn_hour_mode`	Integer	-1	23	11.99	12	4.18	992	1.62%
`after_hours_txn_pct`	Float	0.00	1.00	0.43	0.43	0.14	1,663	2.71%
`after_hours_flag`	Integer	0	1	0.19	0	0.39	49,817	81.12%
`weekend_txn_count`	Integer	0	2,039	30.42	21	32.84	2,157	3.51%
`weekend_txn_pct`	Float	0.00	1.00	0.35	0.33	0.13	2,157	3.51%
`weekend_heavy_flag`	Integer	0	1	0.68	1	0.47	19,643	31.99%
`midnight_txn_count`	Integer	0	467	5.77	4	7.31	8,850	14.41%
`early_morning_txn_count`	Integer	0	225	7.43	5	7.46	6,898	11.23%
`late_night_txn_count`	Integer	0	385	6.69	4	8.29	9,038	14.72%
`midnight_txn_pct`	Float	0.00	1.00	0.07	0.06	0.09	8,850	14.41%
`unusual_hours_flag`	Integer	0	1	0.87	1	0.34	8,070	13.14%
`time_between_txn_mean`	Float	0.00	2194.13	74.00	35.23	139.42	1,660	2.70%
`time_between_txn_median`	Float	0.00	2194.13	57.91	23.38	134.76	1,660	2.70%
`time_between_txn_min`	Float	0.00	2194.13	16.08	0.22	112.30	1,660	2.70%

Feature	Type	Min	Max	Mean	Median	Std Dev	Zero Count	Zero %
time_between_txn_max	Float	0.00	2194.13	230.56	162.46	226.25	1,660	2.70%
time_between_txn_std	Float	0.00	1510.38	61.34	34.58	88.29	2,233	3.64%
very_short_time_between_txn	Integer	0	913	1.33	0	6.21	38,323	62.41%
rapid_fire_txn_count	Integer	0	1,662	9.97	2	25.46	20,785	33.85%
rapid_fire_flag	Integer	0	1	0.25	0	0.43	46,191	75.22%
txn_hour_std	Float	0.00	16.26	5.04	5.09	1.21	1,826	2.97%
consistent_timing_flag	Integer	0	1	0.00	0	0.04	61,333	99.87%
business_hours_txn_pct	Float	0.00	1.00	0.37	0.38	0.12	2,189	3.56%
non_business_hours_heavy	Integer	0	1	0.06	0	0.23	57,822	94.16%
months_active	Integer	0	3	2.89	3	0.47	907	1.48%
txn_per_month	Float	0.00	1663.67	30.80	19.67	33.84	907	1.48%

Validation Checks

☑ No Unexpected Negative Values

☑ No Infinite Values

Outlier Detection (Values > 3 Standard Deviations)

- txn_hour_mode: 907 outliers (1.48%)
- after_hours_txn_pct: 2,473 outliers (4.03%)
- weekend_txn_count: 795 outliers (1.29%)
- weekend_txn_pct: 611 outliers (0.99%)
- midnight_txn_count: 720 outliers (1.17%)
- early_morning_txn_count: 954 outliers (1.55%)
- late_night_txn_count: 797 outliers (1.30%)
- midnight_txn_pct: 725 outliers (1.18%)
- time_between_txn_mean: 1,087 outliers (1.77%)
- time_between_txn_median: 1,060 outliers (1.73%)

... and 11 more features with outliers

Flag Feature Distributions

Feature	True Count	True Percentage	False Count	False Percentage
after_hours_flag	11,593	18.88%	49,817	81.12%
weekend_heavy_flag	41,767	68.01%	19,643	31.99%
unusual_hours_flag	53,340	86.86%	8,070	13.14%
rapid_fire_flag	15,219	24.78%	46,191	75.22%

Feature	True Count	True Percentage	False Count	False Percentage
`consistent_timing_flag`	77	0.13%	61,333	99.87%

Summary

- ☑ **Total Features**: 25
- ☑ **Data Completeness**: 4.00%
- ☑ **Customers Covered**: 61,410
- ☑ **Flag Features**: 5

Report generated automatically by feature engineering pipeline